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Digital twin-driven multi-metrics condition assessment of eccentrically loaded reinforced concrete columns

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ABSTRACT

It is crucial to assess the structural condition of reinforced concrete (RC) structures to ensure their functionality and safety; however, accurate condition assessment remains challenging owing to uncertainties in geometric modelling, finite element analysis and assessment criteria. This study proposes a digital twin-driven multi-metrics method for condition assessment of eccentrically loaded RC columns with cracks. Three RC columns were tested, with one column used for development and two for validation specimens. The geometric digital twin (gDT) of the RC column was constructed using point cloud data for geometry and rebar scanning for reinforcement layout, thereby reducing geometric uncertainties caused by manufacturers. Building on the gDT, a nonlinear finite element model was constructed as the virtual entity and calibrated to experimental data using an optimisation algorithm, thereby producing a mechanical digital twin (mDT) that mitigates analysis-related inconsistencies. The established digital twin was subsequently used to predict the responses of two validation specimens, achieving prediction errors within 5% for ultimate load-bearing capacity. Finally, a multi-metrics condition assessment framework based on DT was proposed, enabling assessment at each loading level for all RC columns. The results indicate that the DT-driven approach substantially enhances the accuracy of structural condition assessment of the RC columns with cracks.

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Reinforced concrete columns; monitoring and inspection; digital twin; multi-metrics condition assessment

1. Introduction

Harsh environment and overloading may cause material deterioration, cracking, deformation and overstress in reinforced concrete (RC) columns of short and medium span bridges, which in turn affect the serviceability and safety of the bridges (Sun et al., 2023; Zaghian et al., 2023). Furthermore, due to uncertainties involved in design and construction, the working conditions of RC columns may not be the same as designed. For instance, ideally axially loaded RC columns may become eccentrically loaded RC columns due to construction defects. It is thus necessary to conduct condition assessment of RC columns from time to time to ensure their serviceability and safety in their life cycles (Jiang et al., 2023).

Inspections and digital image processing techniques are currently utilised to assess the conditions of RC columns, in which the conditions of a cracked RC column are classified mainly by comparing inspected crack characteristics with allowable crack width or crack density limit (Afzali et al., 2023; Hamidia et al., 2023; Ministry of Transport of the People's Republic of China, 2011). However, the allowable crack width or crack density limit is often set up by rules of thumb, and the cracks observed are only an apparent feature on the surface of the RC column. Therefore, this kind of method is difficult to reproduce the actual response of the cracked RC column and provide an accurate condition assessment. Nonlinear finite element analysis (NFEA) is another common way for the condition assessment of RC columns in terms of various parameters such as stiffness, crack, stress and bearing capacity (Abbas et al., 2016; Lin et al., 2017). However, since there are geometric, material and analytical uncertainties involved in NFEA and difficulties in simulating distributed cracks, NFEA often does not produce accurate condition assessments of cracked RC columns.

To solve the problem associated with the uncertainties involved in NFEA, model updating techniques are utilised to reduce uncertainties. Chen et al. (2024) updated the material parameters of both reinforcement and concrete of simply supported RC beams using measured concrete strains. Zheng et al. (2020) carried out nonlinear model updating to reduce uncertainties in NFEA of a cable-stayed bridge using measured natural frequencies and acceleration responses. Xiao et al. (2023) proposed a method to update material parameters of simply supported RC beams based

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on static deflection and crack width using an artificial neural network. Nevertheless, in the above-mentioned model updating, the geometry of the FE model used in the updating process is based on design drawings without updating, and the information (e.g. natural frequencies and acceleration responses) from the sensors of a structural health monitoring system is not integrated with the information (cracks and spalling) from the structural inspection for model updating.

Benefitting from the rapid development of information technology, the digital twin concept (Heng et al., 2024; Sang et al., 2023; Tang et al., 2024; Torzoni et al., 2024) may provide an effective framework for more accurate condition assessment of cracked RC columns. The digital twin consists of four major components: (a) a physical entity and information collected from the physical entity; (b) a physical-informed virtual entity simulating the physical entity; (c) the fusion and interaction between the physical and virtual entities to form the digital twin; and (d) the service and prediction provided by the digital twin. It is worth noting that the digital twin is still in its early stage of development in civil engineering, and its definition varies across disciplines according to specific application requirements. The definition adopted in this study does not require real-time interaction between the digital twin and the physical entity.

Based on this premise in bridge engineering, Lin et al. (2021) proposed a digital twin-based collapse fragility assessment of a long-span cable-stayed bridge under strong earthquakes. Zeng et al. (2025) presented a digital twin-empowered performance evaluation of cracked RC beams. Lao et al. (2024) integrated a multiscale crystal plasticity model with fatigue tests to establish a digital twin for providing probabilistic predictions of micro-crack initiation and propagation in the weld zone of orthotropic steel decks. Liang et al. (2025) developed a digital twin for the dynamics of coupled train-track-bridge systems. However, the digital twin-based condition assessment of eccentrically loaded RC columns in short and medium span bridges has not yet been explored.

There are at least three challenging issues to be addressed for digital twin-based condition assessment of eccentrically loaded RC columns. The first issue is how to construct the geometric digital twin (gDT) of a deteriorated RC column using a 3D laser scanner for geometry (Hinks et al., 2013; Yang et al., 2020) and a rebar scanner for reinforcement position and then convert it to the nonlinear FE model of the column. The second issue is how to convert the nonlinear FE model into the digital twin by mapping the structural inspection and health monitoring information to the nonlinear FE model. The third issue is that the currently used assessment criteria in terms of allowable crack width or crack density limit based on the structural inspection information are different from those in terms of deflection and loading capacity based on the structural health monitoring information. Therefore, the multi-metrics condition assessment by integrating structural health monitoring information with structural inspection information is needed (Xu et al., 2019).

This study thus aims at developing a framework for digital twin-based multi-metrics condition assessment of eccentrically loaded RC columns. The remainder of this paper is organised as follows. Section 2 describes the structural testing and data collection of the eccentrically loaded RC column as a physical entity. Section 3 details the establishment of a gDT of the physical entity. Section 4 establishes a NFEA model of the eccentrically loaded RC column as a virtual entity. Section 5 develops a digital twin of the eccentrically loaded RC column through model updating. Section 6 uses the digital twin to predict the structural spalling and response of the three eccentrically loaded RC columns. Section 7 explores the relationship between structural health monitoring information and structural inspection information and proposes a multi-metrics condition assessment method based on the theory of evidence. Section 8 summarises the research findings and conclusions from this study. Figure 1 outlines the proposed framework of this study.

2. Experimental program: column design, instrumentation and test observations

2.1. Design of test columns

Three types of eccentrically loaded RC columns were designed and tested in this study: a long column with a relatively large eccentricity (C1), a short column with a relatively large eccentricity (C2) and a long column with a relatively small eccentricity (C3). Column C1 is taken as a physical entity to manifest the full process of digital twin formation, while Columns C2 and C3 are used to demonstrate the predictive capability of the established digital twin of Column 1. All columns had a cross-sectional dimension of 250×300 mm with variations in height and loading eccentricity (e_0). Specifically, Columns C1 and C3 had a height of 2200 mm with a loading eccentricity $e_0 = 200$ mm for Column C1 and $e_0 = 80$ mm for Column C3. Columns C1 and C2 shared the same loading eccentricity $e_0 = 200$ mm, but Column C2 had a height of 1200 mm. A rigid plate with a cross-sectional dimension of 250×500 mm was fixed on the top and bottom of each column to facilitate eccentrically loading and prevent end failure.

In addition, supplementary reinforcements were provided at the corbel regions to ensure that failure occurred in the critical region of the column rather than at the specimen ends. The spacing of the stirrup in the middle of column was 150 mm but the spacing of the stirrup at both ends of the columns decreased to 100 mm. The thickness of concrete cover of the columns was set as 20 mm. Figure 2 displays the dimensions and cross-sectional details of the three columns. All three columns were made with the same mix-material ratio and maintained under identical conditions.

The material parameters of concrete and reinforcement were measured separately before the loading test. Concrete cube specimens measuring 100 mm × 100 mm × 100 mm were fabricated alongside the eccentrically loaded RC columns to measure the compressive strength of the concrete. The average of the ultimate compressive strengths of the

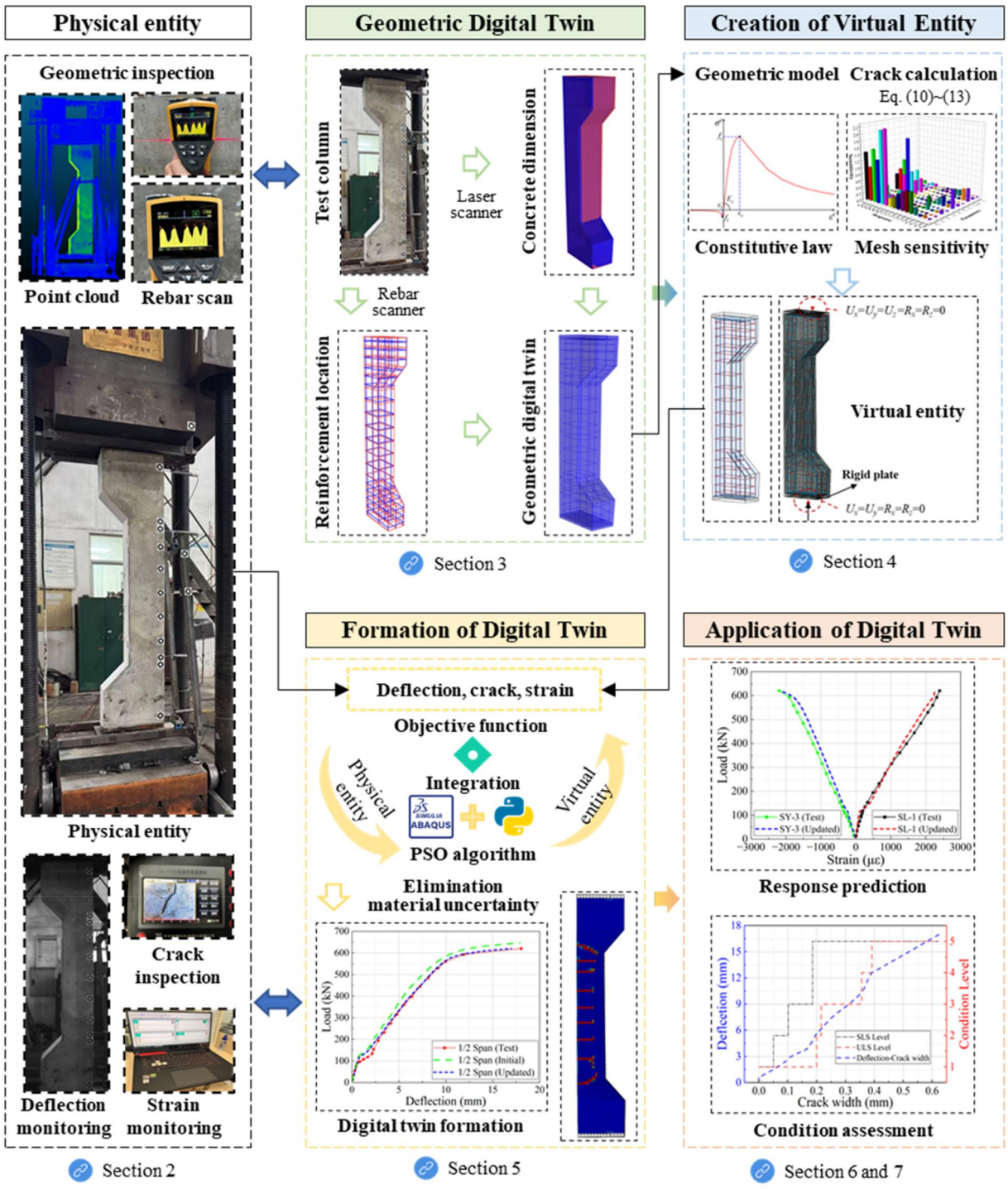


Figure 1. The proposed framework for digital twin-based multi-metrics condition assessment of eccentrically loaded RC columns.

cubic specimens from the same batch was taken as the compressive strength. Concrete tensile strength and elastic modulus were calculated according to fib 2010 code (2010), and Poisson's ratio ν for concrete was assumed to be 0.2, a constant value. The yield strength and ultimate strength of the reinforcement were also tested, and strain gauges were attached to the surface of the reinforcement to measure its elastic modulus. The test material parameters for both reinforcement and concrete are presented in Table 1.

2.2. Experimental setup and instrumentation

Structural inspection and monitoring data were collected separately during the loading test. Inspection data refers to point cloud data (PCD), reinforcement positions, maximum crack width and concrete spalling volume. Monitoring data includes longitudinal reinforcement strains, concrete strains and deflections. The Leica BLK 360 3D laser scanner was used to collect PCD, and the HICHANCE HC-GY62 rebar

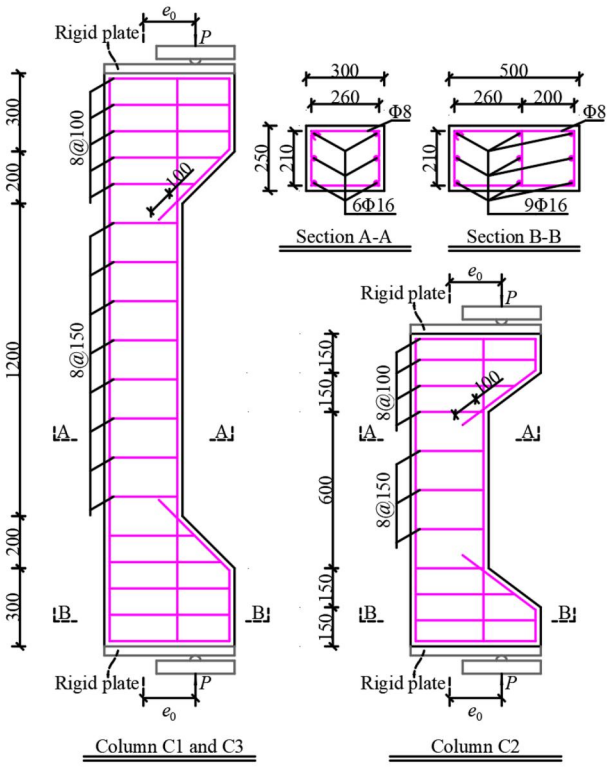


Figure 2. Dimensions and configurations of three eccentrically loaded RC columns (unit: mm).

Table 1. Test material parameters for reinforcement and concrete.

Material	Parameter	Value
Concrete	Compressive strength	f_c = 35.91 MPa
	Young's modulus	E_c = 32,924 MPa
	Poisson's ratio	v = 0.2
	Tensile strength	f_t = 2.76 MPa
Longitudinal reinforcements (16 mm)	Young's modulus	E_s = 212,030 MPa
	Yield strength	f_y = 478.93 MPa
	Ultimate strength	f_u = 647.89 MPa
Stirrups (8 mm)	Young's modulus	E_s = 199,672 MPa
	Yield strength	f_y = 567.62 MPa
	Ultimate strength	f_u = 733.25 MPa

scanner was used to inspect the reinforcement position in the three RC columns. The HIKVISION industrial camera with a 70-megapixel and 50 mm circular targets (mark points) was used for monitoring column deflection. The ZBL-F103 crack width detector was used to inspect concrete maximum crack widths. The Linear Variable Displacement Transducer (LVDT) was placed for monitoring the deflection of the columns and used to check the accuracy of the deflection of the three RC columns at the circular targets measured by the camera. Resistive strain gauges were used to collect concrete and reinforcement strains with grid sizes of 80 mm × 3 mm and 3 mm × 2 mm, respectively. The data acquisition system used was TEST 3826E static strain gauge system. Two laptop computers were used to connect the monitoring acquisition system and the camera.

Figure 3 presents the test setup and the equipment, while Figure 4 displays the placements of the LVDTs, strain gauges, and circular targets of Column C1. Before the test, a preliminary finite element analysis (FEA) was conducted to predict the deformation, load bearing capacity (P_u) and the

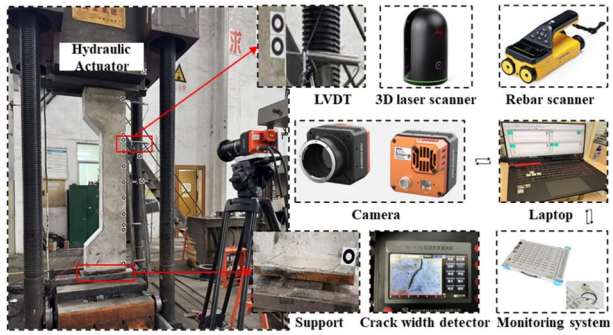


Figure 3. Test setup and equipment.

loading levels for the test. The sensor locations were then determined based on the FEA results. Specifically, the sensor placement was required to capture the column's representative responses.

The testing process was divided into three stages: inspection before loading, inspection and monitoring during loading and inspection after loading. Before loading, the geometry and reinforcement position of the three RC columns were inspected using the Leica BLK 360 3D laser scanner and HICHANCE HC-GY62. These inspection results were used to guide the establishment of gDT. The loading process was controlled by a hydraulic loading system, applying monotonic loading until the RC column failure. The ZBL-F103 crack width detector was employed to inspect crack widths on the tension side of the column during the loading process. The TEST 3826E system and the HIKVISION industrial camera were utilised to measure strains and deflections of the RC columns.

Before cracking during the loading process, the load was increased in increments of 20% of the calculated cracking load. When the load approached the calculated cracking load, the increments were reduced to 5% of the calculated cracking load. After cracking during the loading process, the load was increased in 5% increments of the calculated ultimate load until failure. The calculated cracking and ultimate loads were calculated based on design parameters, which are 122 kN and 628 kN for Column C1, respectively. The loading test was terminated when either of the following conditions was satisfied: (1) structural damage (e.g. the collapse of concrete in the compression zone accompanied by buckling of the reinforcement); (2) the lateral deflection of the column increased rapidly but the loading decreased (e.g. the ultimate load carrying capacity reached). After the loading process, the geometry of the column was re-inspected using the Leica BLK 360 3D laser scanner to quantify the concrete spalling volume.

2.3. Test observations and data collection

The measurement of lateral deflections of the eccentrically loaded column by LVDTs could be affected by crack development. Computer vision (CV) technology using the industrial camera for deflection measurement (Ye et al., 2016) was used in this study. The images taken on the spot were first preprocessed. The sub-pixel edge extraction was then

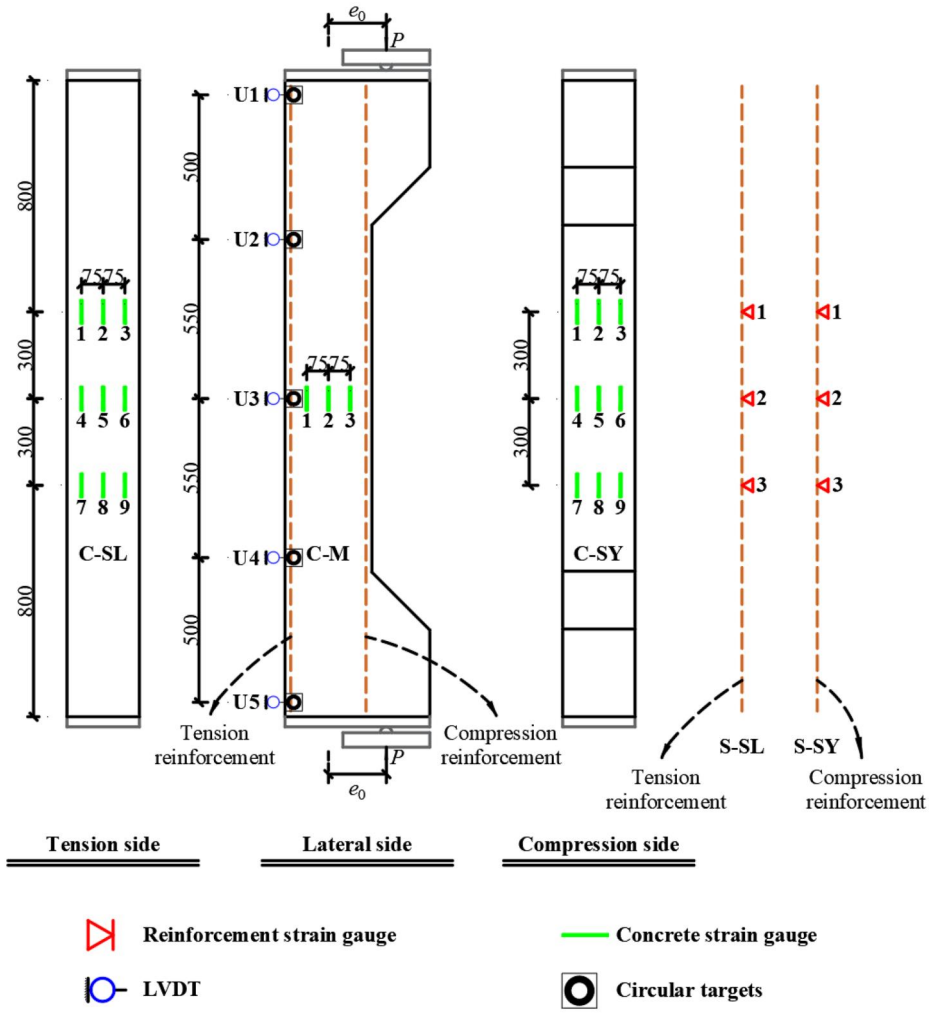


Figure 4. Schematic diagram of placements of monitoring sensors.

carried out to obtain the sub-pixel edge of a circular mark point. The centre of the ellipse was obtained, and the actual deflection was finally calculated from the pixel deflection. Figure 5(a) shows the lateral deflection increments of Column C1 at U3 position measured by the LVDT and the CV. It can be observed that the deflection increments measured by the CV are almost the same as those by the LVDT, and therefore the deflections measured by the CV are used in the subsequent analysis.

It is noted that CV can also capture torsional responses, and its implementation typically follows one of two approaches: (1) moving a single camera to different positions on the structure and analysing the data *via* post-processing to obtain the torsion (Chen et al., 2023), or (2) setting a multi-camera system to capture 3D displacements from different angles, which provides a more direct and robust measurement (Wang & Feng, 2023). While a multi-camera CV system can comprehensively capture complex responses like torsion, it entails higher equipment and computational costs. Therefore, in practice, a hybrid scheme is often preferable. A limited number of cameras for key measurements, supplemented by traditional LVDTs, offers a cost-effective and efficient solution for complex structural responses. Figure 5(b) presents the lateral deflections at U1, U2, U3, U4 and U5 points of the column under various loading levels. Figure 5(c) illustrates the development of the maximum crack width on the tension side of Column C1 under various loading levels, and the maximum crack width occurred near the mid-span of the column (see Figure 7).

Figure 6(a–c) illustrates the development of concrete strains in the tension, compression and lateral sides of Column C1 under various loading levels. It is worth noting that some strain gauges were damaged due to crack development, and only the usable and valid measured strains are shown in the figures. It can be observed from Figure 6(a) that before cracking, concrete strains on the tension side of Column C1 vary linearly and positively as the load increases. However, after cracking, these strains become nonlinear and negative as the load increases. Clearly, the cracks cause the redistribution of concrete strains nearby. Figure 6(b) shows the concrete strains on the compression side, and these compressive strains increase as the load increases. When these compressive strains reach the ultimate concrete strain of 0.0033, the concrete nearby is regarded to reach its ultimate bearing capacity.

Figure 6(c) depicts the concrete strains on the lateral side of the column. Clearly, the neutral axis of Column C1 at the middle section changes as the load increases.

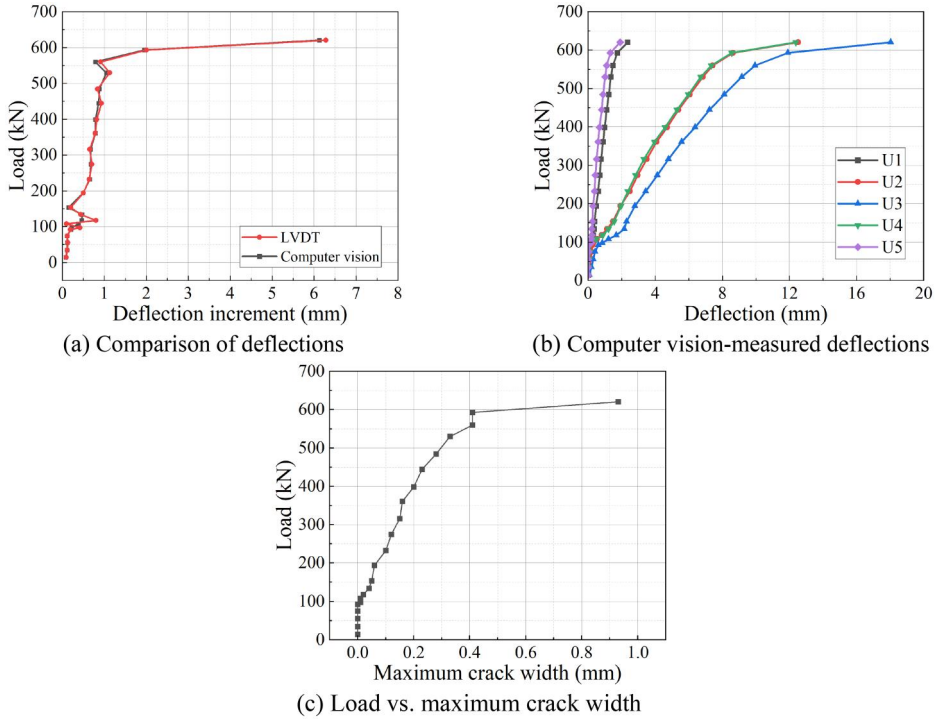


Figure 5. Test results of lateral deflections and maximum crack width.

Specifically, the neutral axis of the column shifts rapidly towards the compression side as the load increases. Figure 6(d) displays the reinforcement strains on the tension and compression sides of Column C1. The reinforcement on the tension side yields before that on the compression side, which aligns with the expected behaviour for large eccentric compression.

The failure mode of Column C1 is tensile failure: the reinforcement on the tension side of the column yields. Figure 7 displays the geometric appearance of Column C1 with damage characteristics after failure. In addition to concrete cracks, the concrete spalling on the compression side of the column can be observed.

3. Establishment of a geometric digital twin

The establishment of a gDT is the first step in the establishment of a digital twin for Column C1. This is because uncertainties during construction lead to discrepancies between the fabricated and designed RC columns in the geometry and reinforcement position. This section details the use of data from the laser scanner and rebar scanner to determine the geometry and reinforcement position and establish a gDT for Column C1.

3.1. Geometry of Column C1

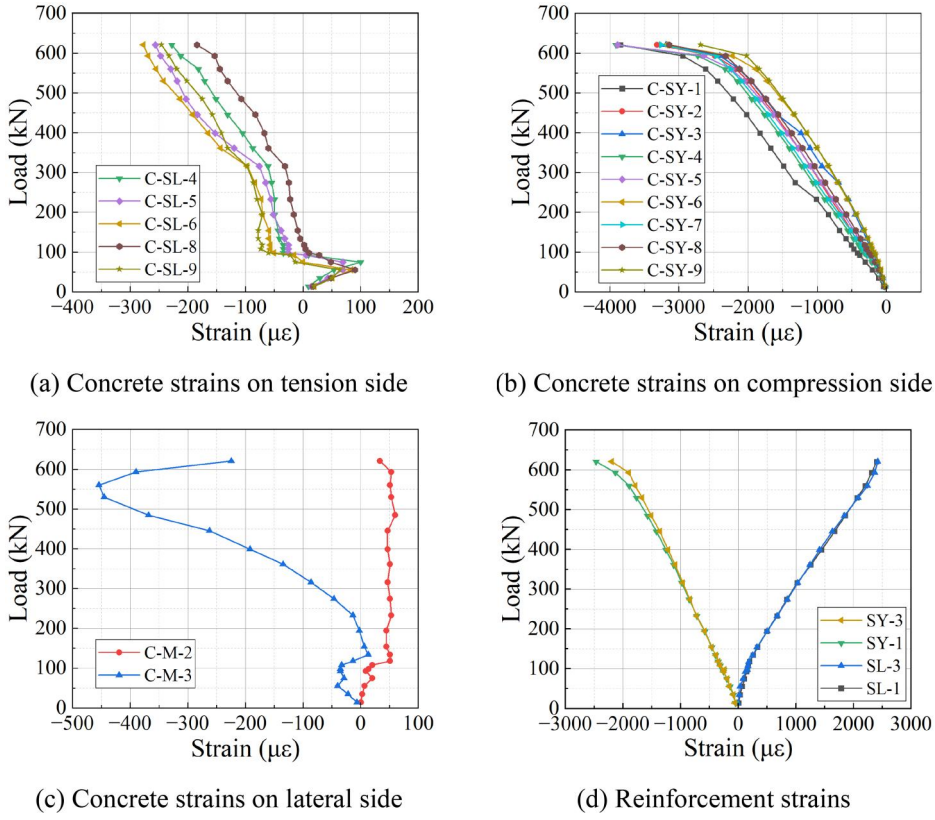
The geometry of Column C1 is derived from the PCD obtained by the 3D laser scanner before the loading test. The PCD collected is pre-processed using CloudCompare 2.6 (Girardeau-Montaut, 2025), which includes the manual cutting of the initial PCD and the 5 mm voxel down-sampling to reduce the data size. The PCD is then rotated by the principal component analysis to align with the appropriate coordinate directions. Column C1 is seen as a geometric object formed by 16 vertex coordinates, which can be considered the gradient of the sample but neglects the flatness of the concrete surface. The vertex coordinates are determined from the equations of the three adjacent planes (Hinks et al., 2013), while the equations of the three adjacent planes are determined by the least squares method based on the local PCD. The algorithm can be seen in Appendix A. The geometry of Column C1 is finally created in Rhino 7.0 (Robert McNeel & Associates, 2010) after the coordinates of the 16 vertices are determined. The process of building the geometry of Column C1 from the PCD is illustrated in Figure 8.

The mean absolute percentage error (MAPE), expressed by Equation (1), is used to quantify the discrepancies in the geometry between the fabricated and designed Column C1:

$$\text{MAPE} = \left(\frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \bar{y}_i}{y_i}\right|\right) \times 100\% \quad (1)$$

where n is the number of values in one geometric quantity; y_i and $\bar{y}_i$ represent the ith design and inspection values in one geometric quantity. The geometry quantity includes height coordinates, width coordinates, length coordinates, reinforcement spacing and cover thickness.

The MAPE values in height and length are 3.22% and 3.48%, and the average error is 2.39 mm and 8.54 mm, respectively. The MAPE value in width is 3.77%, and the average error is 9.44 mm, respectively. It is believed that a higher average width error is due to the unevenness of the floor and the expansion of the concrete when the concrete



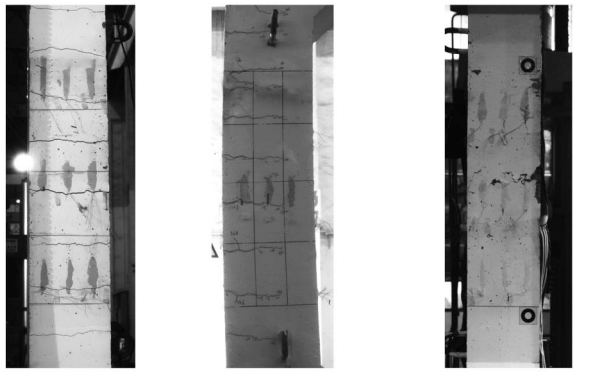
(a) Concrete strains on tension side

(b) Concrete strains on compression side

(c) Concrete strains on lateral side

(d) Reinforcement strains

Figure 6. Test results of concrete strains and reinforcement strains of Column C1.



(a) Tension side (b) Lateral side (c) Compression side

Figure 7. Damage characteristics of Column C1 after failure.

was poured in the width direction during the construction. These errors quantify the discrepancy between the as-built geometry obtained from point cloud data and the design geometry from drawings. This level of error is acceptable for gDT development, since the purpose of a gDT is to faithfully represent the real-world structure rather than reproduce the ideal design geometry. The measured errors, including the slightly larger width deviation attributable to construction factors, are therefore not merely inaccuracies but a valid record of the as-built condition. The overall accuracy (within 5 mm) is consistent with findings in the literature; for example, Zhang et al. (2024) reported ±5 mm differences for a 2,400 mm simply supported beam, which falls within the same range.

3.2. Reinforcement positions

The stirrup spacing and longitudinal reinforcement spacing in the fabricated column may deviate from the design specifications. The rebar scanner can detect the position, depth and distribution of reinforcement by electromagnetic induction. In this study, the stirrup spacing, the longitudinal reinforcement spacing and the concrete cover thickness of the longitudinal reinforcement were inspected using the rebar scanner.

Figure 9(a) describes the reinforcement positions and the measurement positions, and Figure 9(b) presents the comparison of the reinforcement positions between the inspection and the design. Table 2 lists the inspection results of the cover thickness and spacing of the compression and tension reinforcements at the 2 inspection positions as shown in Figure 9(a). It is noted that there are 4 spacings between 3 longitudinal reinforcements with respect to the front and back sides of the column in the width direction.

In Table 3, the inspection results of the stirrup spacing are given. There is a total of 19 spacings among the 18 stirrups with respect to the top and bottom of the column in the height direction. The inspection results show that the mean absolute error of longitudinal reinforcement spacing, cover thickness and stirrups spacing is 21.50 mm, 14.33 mm and 15.6 mm, respectively. There are considerable discrepancies in the longitudinal reinforcement spacing and the cover thickness between the designed and fabricated columns because of the low-quality fabrication. It is thus necessary to establish a gDT of the fabricated column by considering the measured reinforcement positions in the gDT in Rhino.

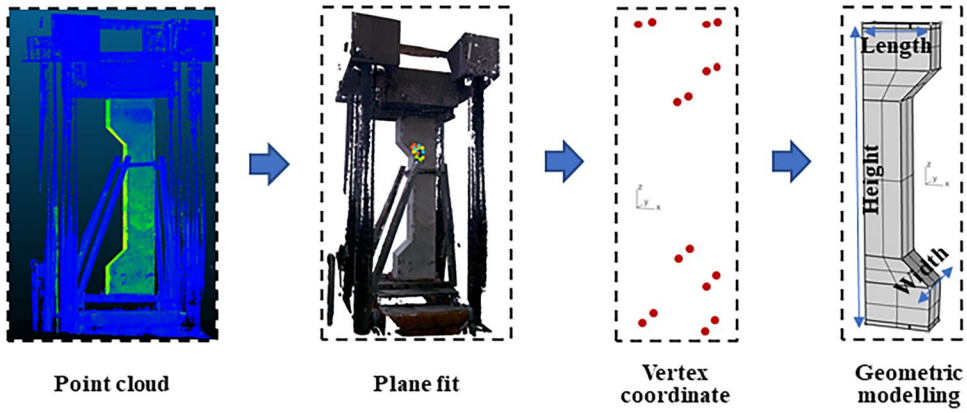
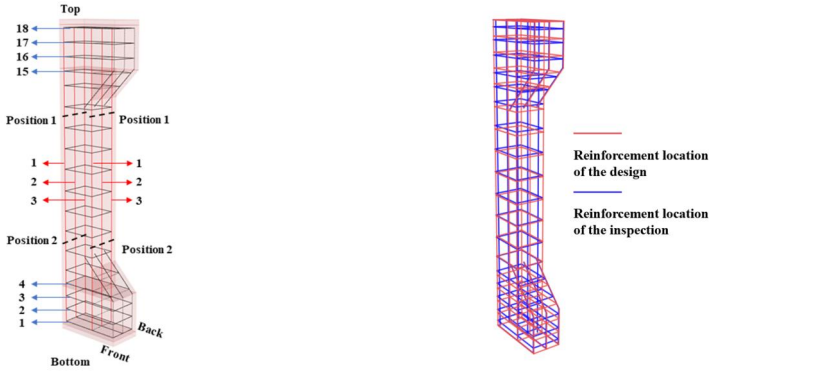


Figure 8. The process for determining geometry of Column C1 from PCD.



(a) Description of reinforcement positions (b) Comparison between inspection and design

Figure 9. Description and comparison of reinforcement positions.

4. Establishment of virtual entity

The gDT of the column is imported into ABAQUS *via* Rhino with IGS format set by the Initial Graphics Exchange Specification (IGES) (Nasr & Kamrani, 2007). The NFEA model of the column is then implemented using ABAQUS 2024 (Dassault Systèmes, 2024). The NFEA model uses C3D8R solid elements for concrete and T3D2 truss elements for all reinforcements. C3D8R solid elements and T3D2 truss elements have only one integration point located in the centre of the element. The stirrups and supplementary reinforcements are modelled using pre-built reinforcement by assuming a complete bond with the concrete. The bond-slip between the longitudinal reinforcements and the concrete is simulated by nonlinear springs (Liu & Bai, 2019).

The boundary conditions of the NFEA model are consistent with the physical entity with the top of the column fixed and the bottom of the column loaded. Thus, the translational degrees of freedom (DOFs) at the bottom of the NFEA model in the x- and y- directions (U_x, U_y) are constrained but those in the z-direction are free for loading. The translational DOFs at the top of the NFEA model in the three directions (U_x, U_y, U_z) are all constrained. The rotational DOFs at both top and bottom of the NFEA model around the x and z axes (R_x, R_z) are constrained. A concentrated force is applied to the rigid plate in the z-direction. The NFEA model generated based on the gDT is taken as a virtual entity of Column C1, as shown in Figure 10.

The concrete damage plasticity (CDP) model (Zhang et al., 2021) is adopted in this study as a concrete constitutive law to describe the nonlinear properties of concrete. The relation between the uniaxial load stress σ and strain ε of concrete in the CDP model is defined by Equations (2)–(7) and plotted in Figure 11(a):

$$\sigma_c = (1-d_c)E_c\varepsilon, \sigma_t = (1-d_t)E_c\varepsilon \tag{2}$$

$$d_c = \begin{cases} 1-\dfrac{\rho_c\beta}{\beta-1+x_c^{\beta}}, & 0 < x_c \le 1 \\ 1-\dfrac{\rho_c}{\alpha_c(x_c-1)^2+x_c}, & x_c > 1 \end{cases},$$

$$d_t = \begin{cases} 1-\rho_t(1.2-0.2x_t^5), & x_t \le 1 \\ 1-\dfrac{\rho_t}{\alpha_t(x_t-1)^{1.7}+x_t}, & x_t > 1 \end{cases} \tag{3}$$

$$\rho_c = \frac{f_c}{E_c\varepsilon_c}, \rho_t = \frac{f_t}{E_c\varepsilon_t} \tag{4}$$

$$\beta = \frac{E_c\varepsilon_c}{E_c\varepsilon_c - f_c} \tag{5}$$

$$x_c = \frac{\varepsilon}{\varepsilon_c}, x_t = \frac{\varepsilon}{\varepsilon_t} \tag{6}$$

$$\alpha_t = 0.312 \times f_t^2 + 0.3, \alpha_c = 0.157 \times f_c^{0.785} - 0.905 \tag{7}$$

Table 2. Inspection results of longitudinal reinforcement spacings and cover thickness.

		Spacing (mm)			Cover thickness (mm)		
Side	Location	Position 1	Position 2	Design	Position 1	Position 2	Design
Longitudinal reinforcement on tension side	Front–No.1	41	43	20	/	/	/
	No.1–No. 2	78	80	105	33	32	20
	No.2–No. 3	84	82	105	33	43	20
	No.3–Back	47	45	20	36	42	20
Longitudinal reinforcement on compression side	Front–No. 1	40	44	20	/	/	/
	No.1–No.2	86	84	105	34	25	20
	No.2–No. 3	86	82	105	33	22	20
	No.3–Back	38	40	20	37	23	20

Table 3. Inspection results of stirrup spacings.

	Spacing (mm)			Spacing (mm)	
Location	Inspection	Design	Location	Inspection	Design
Bottom–No.1	58	25	No.10–No.11	157	150
No.1–No.2	88	100	No.11–No.12	160	150
No.2–No.3	106	100	No.12–No.13	159	150
No.3–No.4	104	100	No.13–No.14	97	150
No.4–No.5	96	100	No.14–No. 15	90	100
No.5–No.6	112	150	No.15–No. 16	112	100
No.6–No 7	145	150	No.16–No. 17	96	100
No.7–No.8	164	150	No.17–No. 18	80	100
No.8–No.9	157	150	No.18–Top	63	25
No.9–No.10	156	150			

where f_c and f_t are the uniaxial compressive and tensile strength of concrete; σ_c and σ_t are the compressive and tensile stress of concrete; ε_c and ε_t are the peak compressive and tensile strain; d_c and d_t are the damage variables for compressive and tensile; α_c and α_t are the parameters of the descending section of the uniaxial compressive and tensile stress–strain curves and the values are 1.70 and 2.67, respectively (Zhang et al., 2021).

In ABAQUS, the CDP model also defines the dilation angle (φ), flow potential eccentricity (e), the ratio of biaxial to uniaxial compressive stresses (f_{br}/f_{cr}) and the ratio of the second stress invariant on the tensile to the compressive meridian (k) to represent the stress-strain relations of the concrete material in the general 3D stress state. In this study, the values of these parameters are selected according to the studies of Hamoda et al. (2023) and Song et al. (2024). Specifically, the values of φ, e, f_{br}/f_{cr} and k are taken as 30°, 0.1, 1.16 and 0.667, respectively. In addition, as the viscosity relaxation parameter (μ) that affects the convergence performance of the FE model, after multiple attempts, it is selected as 0.005, which is consistent with the previous studies (Song et al., 2024).

The double broken line is used to present the stress-strain curve of reinforcement, in which strain hardening is considered. The relation between the stress σ^s and strain ε^s for reinforcement is defined by Equation (8) and plotted in Figure 11(b). The relation between the bond shear stress τ_b and slip s is defined by Equations (9)-(10) according to the fib 2010 code (2010) and shown in Figure 11(c):

$$\sigma^s = \begin{cases} E_s\varepsilon^s, & \varepsilon^s < \varepsilon^{s1} \\ E_s\varepsilon^{s1} + E_i(\varepsilon^s - \varepsilon^{s1}), & \varepsilon^s \geq \varepsilon^{s1} \end{cases} \quad (8)$$

$$\tau_b = \tau_{b\max}(s/s_1)^{\alpha}, 0 \leq s \leq s_1 \quad (9)$$

$$\tau_{b\max} = 2.5\sqrt{f_c}, \tau_{bf} = 0.4\tau_{b\max}, \quad (10)$$

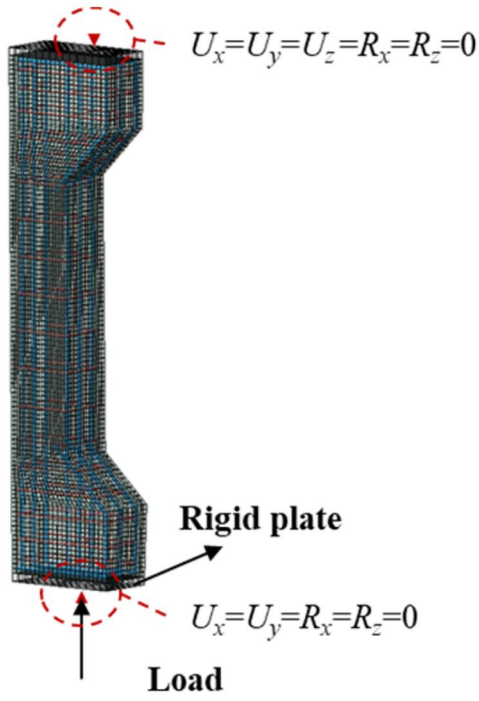
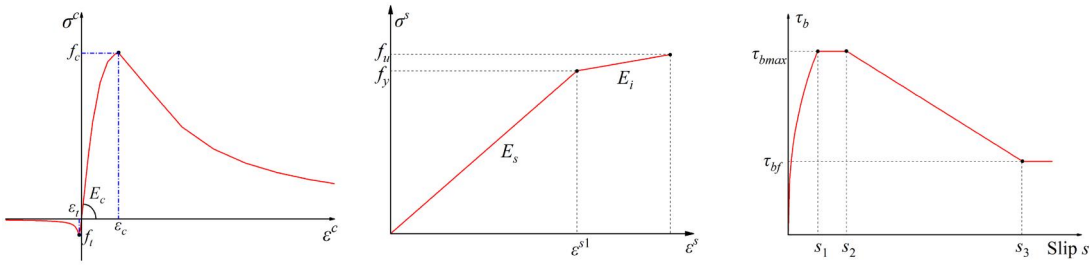


Figure 10. Virtual entity of Column C1.

where ε^{s1} is the yield strain of the reinforcement; E_i is the slope of the reinforcement after yielding; $\tau_{b\max}$ is the peak bond shear stress; τ_{bf} is the residual bond shear stress; s_1, s_2 and s_3 are 1, 2 and 8 mm as defined in fib 2010 code (2010), respectively.

Since the cracks of the column developed as the load increased and the gDT of the column was established prior to the load testing, the cracks of the column in the virtual model are thus calculated and treated as one kind of structural response in addition to strain and deflection. The crack width of the column is calculated based on the CDP model *via* tension damage and concrete strain (Zhang et al., 2021). The deformation of the crack is diffused into the total deformation of the element and expressed as an average strain. For a C3D8R element with a crack, the total strain ε of the element is assumed to decompose into the elastic strain ε_e and the crack strain ε_f of the element (Ju, 1989), as expressed by Equation (11). The elastic strain ε_e can be expressed by Equation (12) according to Equation (2), and the crack strain ε_f can then be deduced and expressed by Equation (13). The crack width w_f in the element can be finally determined by Equation (14):



(a) Concrete constitutive law (b) Reinforcement constitutive law (c) Bond-slip constitutive law

Figure 11. Constitutive laws of concrete and reinforcement.

Table 4. The calculation time and error with test for different mesh sizes.

	Deflection (mm)		The maximum crack width (mm)		
Mesh size (mm)	RMSE	MAE	RMSE	MAE	Calculation time (min)
10	0.7227	0.5088	0.1265	0.0928	954
20	0.8308	0.5128	0.1331	0.0968	31
30	0.8923	0.7134	0.1523	0.1129	20
40	0.9449	0.8374	0.1909	0.1434	15

$$\varepsilon = \varepsilon_e + \varepsilon_f \tag{11}$$

$$\varepsilon_e = \frac{\sigma}{E_0} = \frac{(1-d_t)E_0\varepsilon}{E_0} = (1-d_t)\varepsilon \tag{12}$$

$$\varepsilon_f = \varepsilon - \varepsilon_e = \varepsilon - (1-d_t)\varepsilon = d_t\varepsilon \tag{13}$$

$$w_f = \varepsilon_f l_E = d_t \varepsilon l_E \tag{14}$$

where l_E is the equivalent length of the element and it is calculated by $l_E = m_d{}^{1/n_d}$, in which m_d is the volume of the element and n_d is the dimension of the element taking 3 for C3D8R element.

It should be noted that the accuracy of crack width calculated by Equation (14) depends on the size of the element. A mesh sensitivity study comparing sizes from 10 mm to 40 mm showed that the 20 mm mesh provides a favourable balance between accuracy and computational cost, yielding errors comparable to the 10 mm mesh but with a significantly shorter simulation time. The mid-span deflection and maximum crack width at the test loading levels were compared; errors were quantified by root mean square error (RMSE) and mean absolute error (MAE). A quantitative comparison of errors and computational times for different mesh sizes is provided in Table 4. In this study, the maximum crack width is defined as the maximum value of the crack widths in all the elements of the virtual model.

The established virtual model is now used to calculate the deflection, strain and maximum crack width of the column under loading. Figure 12 displays the variations of deflection, strain and maximum crack width of the column with loading, obtained from both the virtual model and the loading test. It can be observed that the deflection and concrete strain obtained from the virtual model are smaller than the test results. The maximum crack width obtained from the virtual model, on the other hand, is higher than the test results.

According to Lu et al. (1989), the ultimate load-bearing capacity P_u of the column can be determined by the intersection point between the reference line constructed with a slope equal to three times the initial elastic stiffness (tan $\psi = 3\tan\theta$ in Figure 12(a)) and the load-deflection curve calculated by the virtual entity. As a result, the ultimate load bearing capacity P_u calculated by the virtual entity is 646.8 kN, compared with 620.3 kN from the loading test. The discrepancies between the calculated and test results are mainly due to the uncertainties in the material properties and the assumptions in the constitutive laws. Therefore, it is necessary to convert the virtual entity into a digital twin of the physical entity by mapping the test results from the physical entity to the virtual entity.

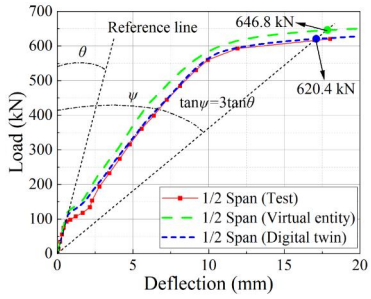
5. Formation of digital twin

In this section, the test data collected from the column is mapped to the virtual entity using the particle swarm optimisation (PSO) algorithm to form a digital twin of the column.

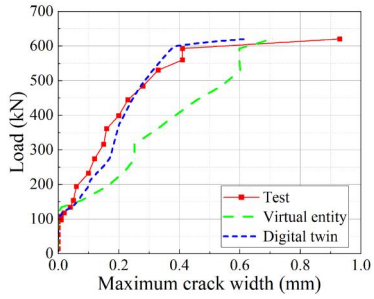
5.1. Updating parameters

Since the virtual model of the column was established based on the gDT of the column, the geometric uncertainties should not be considered, and only the uncertainties in the material properties and the constitutive laws should be considered in the virtual model updating. To this end, a sensitivity study of the deflection (U3), concrete strain (C-SY-5) and maximum crack width to all the candidate parameters is performed under different load levels.

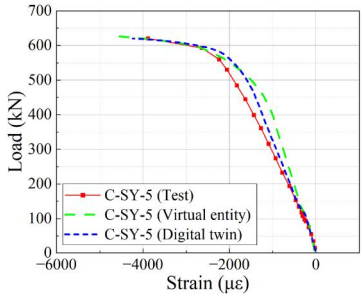
In consideration of life cycle performance assessment of the column, the selected load levels include 4 pre-yield levels (194.1, 273.9, 360.8 and 444.8 kN) and 2 post-yield levels (530 kN and 560 kN), as shown in Figure 13(a). However, the sensitivity analysis of the maximum crack width excludes the post-yield load levels because the crack width already exceeds the permissible threshold of 0.2 mm at these load levels. To calculate the sensitivity of the parameters, a



(a) Load-deflection curves

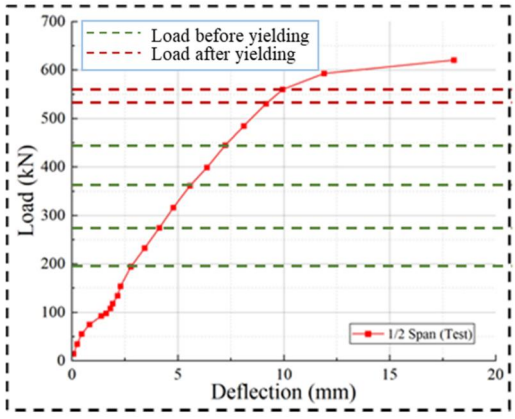


(b) Load-maximum crack width curves

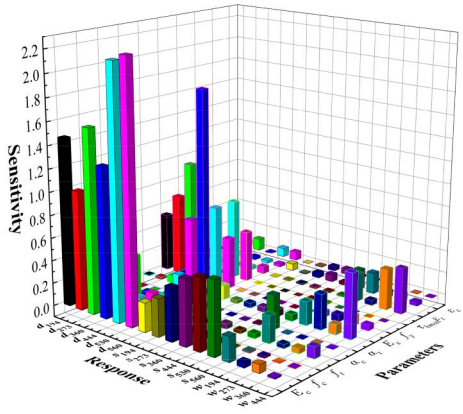


(c) Load-concrete strain curves for compression side.

Figure 12. The column responses obtained from the virtual entity, test and digital twin.



(a) Load levels for sensitivity analysis and updating



(b) The results of sensitivity analysis

Figure 13. Load levels, sensitivity analysis and updating parameters.

variation of 10% is applied to each parameter, and the corresponding changes of the considered responses are calculated.

The sensitivity is defined by the ratio between the changes of the parameters and the responses, as per standard practice suggested by Gou et al. (2022). A total of ten material parameters (E_c, f_t, f_c, α_t, α_c, ε_t, ε_c, E_s, f_y, $\tau_{b\max}$) are selected in sensitivity analysis, and the results of sensitivity analysis are presented in Figure 13(b). The results show that the five parameters E_c, f_t, α_t, E_s and $\tau_{b\max}$ are sensitive to the column responses before reinforcement yielding, and the three parameters f_c, α_c and f_y are sensitive to the column responses after reinforcement yielding. The parameters ε_c and ε_t are not sensitive to the column responses, and therefore they are not included in the updating parameters.

5.2. Objective functions and optimisation algorithm

In consideration of the results of sensitivity analysis, the virtual model updating is divided into two stages. In the first updating stage before reinforcement yielding, the five updating parameters are E_c, f_t, α_t, E_s and τ_{bmax}, and the objective function expressed by Equation (15) includes the deflection f_d of the column at 1/4, 1/2 and 3/4 spans, the concrete strain f_s of the column under 194.1, 273.9, 360.8 and 444.8 kN, and the maximum crack width f_w. It is noted that only the concrete strains on the compression side are involved in the objective function, while the other concrete strains and reinforcement strains are used to validate the quality of the digital twin. In the second updating stage after reinforcement yielding, the three updating parameters are f_c,

α_c and f_y, and the objective function expressed by Equation (16) includes the deflections of the column at the 1/4, 1/2 and 3/4 span points and the concrete strain of the column under 530 kN and 560 kN. It is noted that the parameters obtained from the first stage updating remain unchanged when the second stage updating is performed:

$$\min J_{\mathrm{OF1}}(x) = \min\{f_d(x), f_s(x), f_w(x)\} \tag{15}$$

$$\min J_{\mathrm{OF2}}(x) = \min\{f_d(x), f_s(x)\} \tag{16}$$

$$\begin{aligned} f_d\ (x) &= \sum_{j=1}^{m_d}\sum_{i=1}^{n_d}\left(\frac{d_{\mathrm{VE},i,j} - d_{\mathrm{PE},i,j}}{d_{\mathrm{PE},i,j}}\right)^2, \\ f_s(x)\ &= \sum_{j=1}^{m_d}\sum_{i=1}^{n_d}\left(\frac{s_{\mathrm{VE},i,j} - s_{\mathrm{PE},i,j}}{s_{\mathrm{PE},i,j}}\right)^2, \\ f_w(x) &= \sum_{j=1}^{m_w}\left(\frac{w_{\mathrm{VE},j} - w_{\mathrm{PE},j}}{w_{\mathrm{PE}j}}\right) \end{aligned} \tag{17}$$

where $J_{\mathrm{OF1}}(x)$ and $J_{\mathrm{OF2}}(x)$ are the objective functions for the first and second updating stage, respectively; n_d and n_s are the number of deflections and concrete strains concerned, respectively; m_d and m_w are the number of load levels for deflection and concrete strain as well as maximum crack width; $d_{\mathrm{VE},i,j}$ and $d_{\mathrm{PE},i,j}$ are the deflections obtained from the virtual entity and the loading test, respectively, under the jth load level and at the ith measurement points; $w_{\mathrm{VE},j}$ and $w_{\mathrm{PE},j}$ are the maximum crack widths obtained from the virtual entity and the inspection of the physical entity under the jth load level, respectively; $s_{\mathrm{VE},i,j}$ and $s_{\mathrm{PE},i,j}$ are the concrete strains on the compression side obtained from the virtual entity and the loading test of the physical entity, respectively, under the jth load level at the ith measurement points.

The PSO algorithm is then used to find the optimal parameters which minimise the two objective functions. According to preliminary simulations, the population size for the mutation-enabled PSO is set to 12 because of computational availability constraints; the maximum number of iterations is 15, and the percentage of particle mutation is 50%, and the iteration process terminates if the best solution remains unchanged for 5 consecutive iterations. The upper and lower bounds for the updating parameters are set at 25% of their initial values. The computations were conducted on 32 cores of an AMD Ryzen 3970X processor. In addition, the workflow for updating in ABAQUS can be seen in Appendix B.

5.3. Formation of digital twin

Table 5 shows the initial values and updated values of the eight parameters concerned for the virtual model updating. The combination of increased $\tau_{b\max}$ and decreased E_s balances the interaction between the reinforcement and concrete. The restrained effect of the stirrups makes the parameter f_c increase, whereas the tensile reinforcement effect makes the parameter α_t decrease. Moreover, the decreased E_c and E_s make the stiffness of the virtual entity decrease to approach the stiffness of the physical entity. The updated virtual model is called the digital twin.

Table 5. Changes in material parameters before and after updating.

Material parameter	Initial value	Updated value	Rate
E_c (MPa)	32,924.00	28,717.00	−12.77%
f_c (MPa)	35.91	41.60	15.84%
f_t (MPa)	2.76	2.83	2.53%
α_c	1.70	1.76	3.53%
α_t	2.67	2.97	11.07%
E_s (MPa)	212,030.00	184,217.00	−13.12%
f_y (MPa)	478.00	423.00	−11.51%
$\tau_{b\max}$ (MPa)	8.71	10.07	13.51%

Figure 12 presents the variations of the deflection, maximum crack width, and concrete strain of the column with loading levels, which are obtained from the loading test, the virtual model and the digital twin, respectively. It can be observed that the results from the digital twin align more closely with the test results than the virtual entity. For instance, the maximum crack widths of the column under the load levels of 232.6 and 399.0 kN calculated by the virtual model are 0.23 mm and 0.26 mm, whereas the maximum crack widths under the same conditions predicted by the digital twin are 0.13 mm and 0.21 mm. Since the maximum crack widths measured from the loading tests are 0.10 mm and 0.20 mm, the results from the digital twin align more closely with the test results than the virtual entity. Moreover, the ultimate load-bearing capacity P_u of the column predicted by the virtual model and the digital twin is 646.8 and 620.4 kN, respectively, while the test result was 620.3 kN. Again, the result from the digital twin aligns more closely with the test result than the virtual entity.

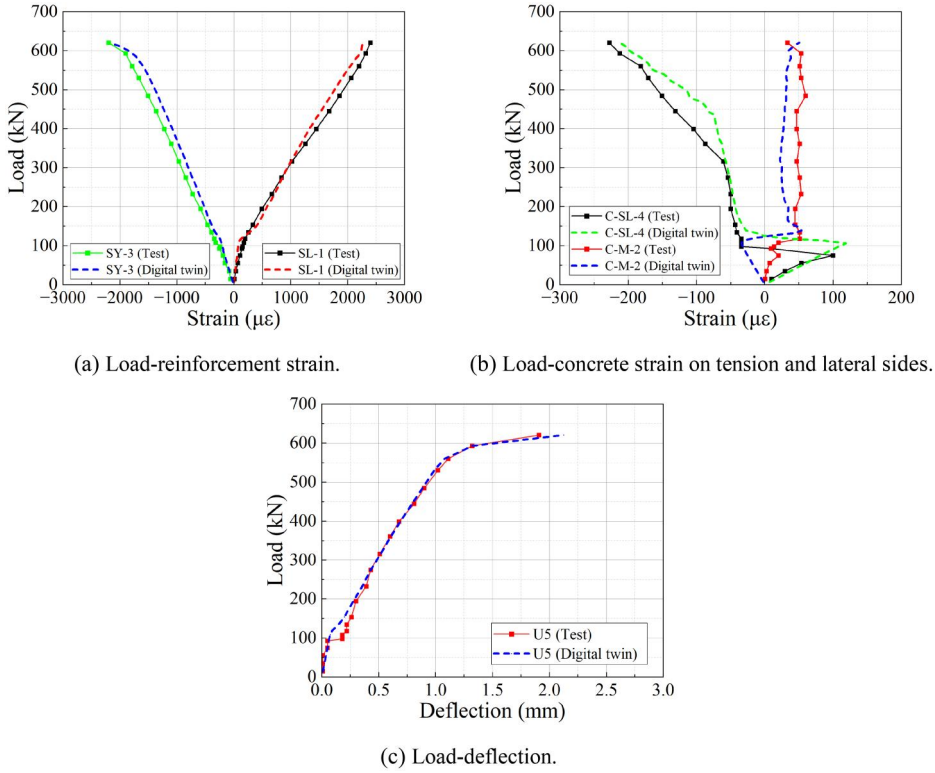
6. Application of digital twin

6.1. Prediction of structural response by digital twin

In the virtual model updating, the concrete strains on the tension and lateral sides, reinforcement strains and support deflections of the column measured from the loading tests are not used in the objective function. Therefore, these structural responses are predicted by the digital twin and compared with the measurement results so as to verify the established digital twin. Figure 14 shows the variations of the reinforcement strain, concrete strains and support deflection with loading levels, obtained from the digital twin and the loading tests. It can be observed that the responses predicted by the digital twin are close to the measurement results, indicating that the digital twin can be utilised to predict the structural responses of the column at the locations without sensors.

6.2. Prediction of concrete spalling by digital twin

The concrete spalling occurred when Column C1 approached its failure. It is difficult to quantify the volume of concrete spalling using traditional methods. However, the PCD obtained from 3D laser scanning after the loading test offers a reliable solution (Zhang et al., 2022). The PCD collected is pre-processed using CloudCompare 2.6 (2025),



(a) Load-reinforcement strain.

(b) Load-concrete strain on tension and lateral sides.

(c) Load-deflection.

Figure 14. The structural responses predicted by the digital twin.

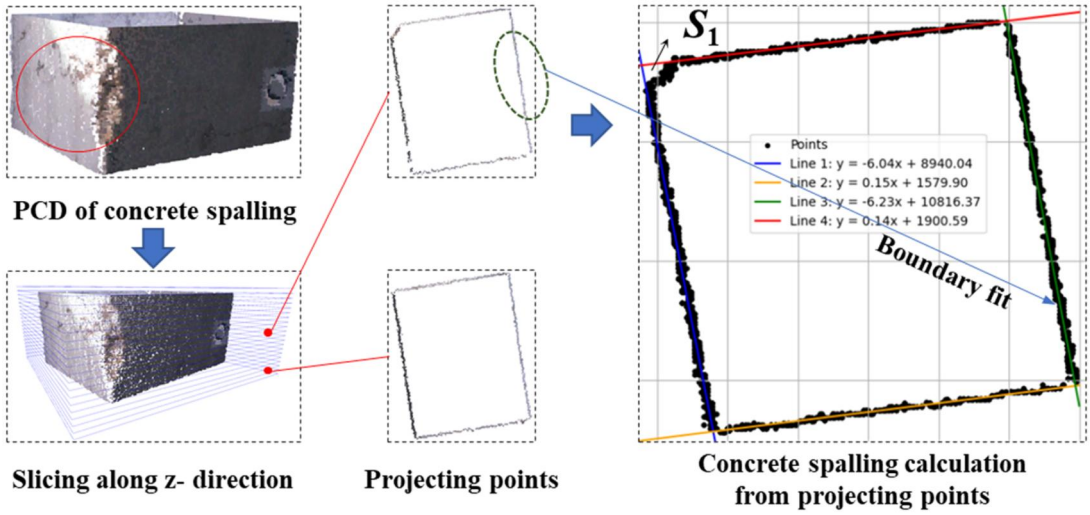
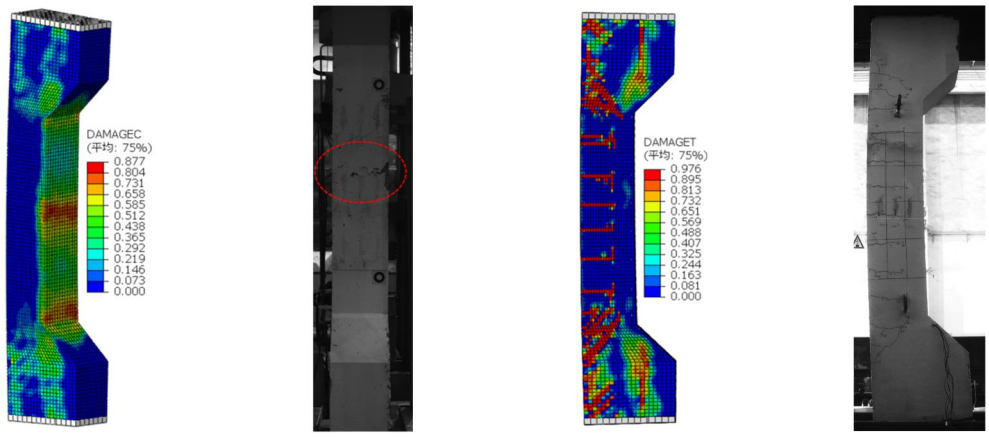


Figure 15. Flowchart of estimating concrete spalling volume from point cloud data.

which includes the manual cutting of the initial PCD and the 5 mm voxel down-sampling to reduce the data size. The PCD is then rotated by the principal component analysis to align with appropriate coordinate directions.

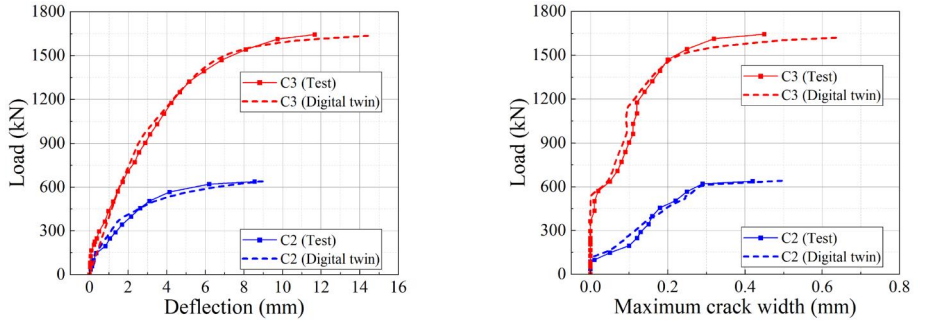
The points of sliced PCD along the *z*- direction in an interval of 5 mm is projected to the x-y plane. The intersection points of the boundary edges fitted by the least squares method are determined. The concrete spalling area S_i is integrated from the intersection point to the spalling points in *i*th sliced PCD. The concrete spalling volume V is finally calculated by the summation of the total concrete spalling area of the sliced PCD. The process of quantifying the concrete spalling volume from the PCD is illustrated in Figure 15. As a result of the above process, the concrete spalling volume of Column C1 is 81.8 cm^3.

In the digital twin, if the compressive damage d_c of the concrete element exceeds the threshold, it is regarded as concrete spalling (Rajput & Iqbal, 2017). The volume of concrete spalling is determined based on the number of elements that exceed the threshold. The compressive damage d_c corresponding to the concrete ultimate compressive strain is considered as the threshold, and the value is 0.72 based on the updated material parameter. When the digital twin reaches its ultimate load bearing capacity, the number of elements with concrete spalling is 19, and the volume of concrete spalling of the digital twin is 106.4 cm^3, compared



(a) Contours of compression damage d_c (b) Compression damage in test (c) Contours of tension damage d_t (d) Tension damage in test

Figure 16. Comparison of damage contours between digital twin and physical entity.



(a) Load-deflection curves (b) Load-maximum crack width curves

Figure 17. Comparison of structural responses of columns C2 and C3 between test and prediction.

with 81.8 cm^3 from the physical entity. Figure 16 compares the contours of tension damage d_t and compression damage d_c between the digital twin and the physical entity. It can be observed that the damage contours from the digital twin match those from the physical entity.

6.3. Prediction of structural behaviour of the other two RC columns

The three RC columns tested in this study are all the RC compression-bending members with identical material properties although their geometric sizes or loading eccentricities are different. Therefore, the updated material properties from the digital twin of Column C1 can be applied to the gDTs of Columns C2 and C3 to predict their structural performance directly without conducting another model updating. Figure 17 displays the comparison of load-deflection curves and load-maximum crack width curves between the test and the prediction.

The predicted ultimate load-bearing capacity P_u of Column C2 is 640.2 kN, while the test result is 638.2 kN. Such a difference is slightly larger than that of Column 1, which may be due to different loading eccentricities. For Column C3, the predicted and test ultimate load bearing capacities P_u are 1638.0 and 1644.3 kN, respectively. Such a difference is also slightly larger than Column 1, which may be due to different column heights. In general, the prediction results match the test results, which means the material parameters obtained from the digital twin of Column 1 can be used to predict the other two columns. The close agreement between the digital twin predictions and the test results demonstrates the robust predictive capability of the proposed digital twin framework.

7. Digital twin-driven condition assessment

7.1. Condition assessment of RC columns according to specifications

Serviceability and safety are two important aspects in the structural condition assessment of RC columns. According to the serviceability limit state (SLS) specified in Chinese Code GB 50292-2015 (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2015), the thresholds of the maximum crack width and deflection of RC bending members are 0.2 mm and $L_0/200$, respectively, where L_0 denotes the member span. For the ultimate limit state (ULS), the corresponding threshold is 0.5 mm for the maximum crack width and $L_0/120$ for deflection of RC bending members. For the bearing capacity of RC bending members in ULS, the threshold is $R/\gamma_0 S \geq 0.95$,

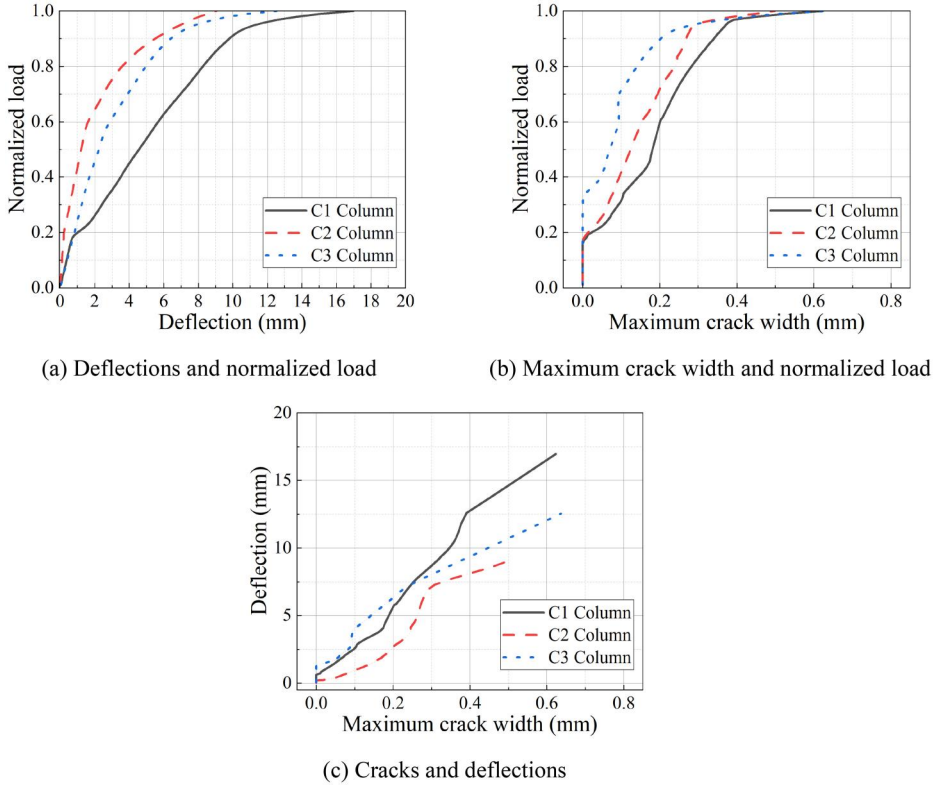
where R, γ_0 and S are the ultimate load bearing capacity P_u, the structural importance factor, and the load on the column, respectively. The structural importance factor γ_0 is taken as 1.1, according to the GB 50010-2010 (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2010).

With reference to the test results presented in Section 2.3 and Figure 5: for Column C1 under a load of 444.8 kN, the mid-span deflection is 7.23 mm, which is far below the SLS threshold of 11 mm, but the maximum crack width is 0.23 mm, which already exceeds the SLS threshold of 0.2 mm. Moreover, when the load is increased to 592.9 kN, the mid-span deflection of the column is 11.9 mm, which just surpasses the SLS threshold of 11 mm, but the maximum crack width is 0.41 mm, which far exceeds the SLS threshold of 0.2 mm. Therefore, the currently used SLS thresholds lead to contradictory assessment results. This phenomenon may be related to the different focuses of the maximum crack width and deflection metrics in the specification, the former reflecting the local characteristics of the structure, while the latter reflecting the overall behaviour of the structure. Nevertheless, it is difficult to assess the SLS conditions of the column according to the specification because the assessment results according to the two SLS thresholds provided by the specification are not consistent.

Figure 18(a) presents the relations between the mid-span deflection of the three columns and the normalised load, obtained or predicted by the digital twin. The normalised load herein is defined as the load divided by the ultimate load bearing capacity P_u. Figure 18(b) displays the curves of the maximum crack widths of the three columns and the normalised load, obtained or predicted by the digital twin. Based on Figure 18(a,b), one can obtain the relations between the mid-span deflections and the maximum crack widths of the three columns, as shown in Figure 18(c). Clearly, the digital twin can provide more detailed information for structural condition assessment.

Figure 19 displays the condition assessment results of the three RC columns according to the two SLS thresholds and the three ULS thresholds provided by the Chinese Code GB 50292-2015 (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2015) in terms of the results obtained by the digital twin and presented in Figure 18. The condition assessment results of the three RC columns are clearly inconsistent. For instance, for the SLS assessment of Column C1, when the normalised load is 0.61, the column reaches the SLS threshold of the maximum crack width, but when the normalised load becomes 0.96, the column approaches the SLS threshold of the deflection.

For the ULS assessment of Column C1, when the normalised load is 0.96, both the maximum crack width and mid-span deflection of the column are smaller than their respective thresholds, but the column already reaches the ULS threshold of the ultimate load-bearing capacity. Similar observations can be made for the ULS assessment of Column C2. Therefore, it is necessary to conduct multi-source information fusion for condition assessment of RC columns by leveraging the digital twin and the Dempster-Shafer (DS) evidence theory.



(a) Deflections and normalized load

(b) Maximum crack width and normalized load

(c) Cracks and deflections

Figure 18. Relationship between monitoring and inspection data based on digital twins.

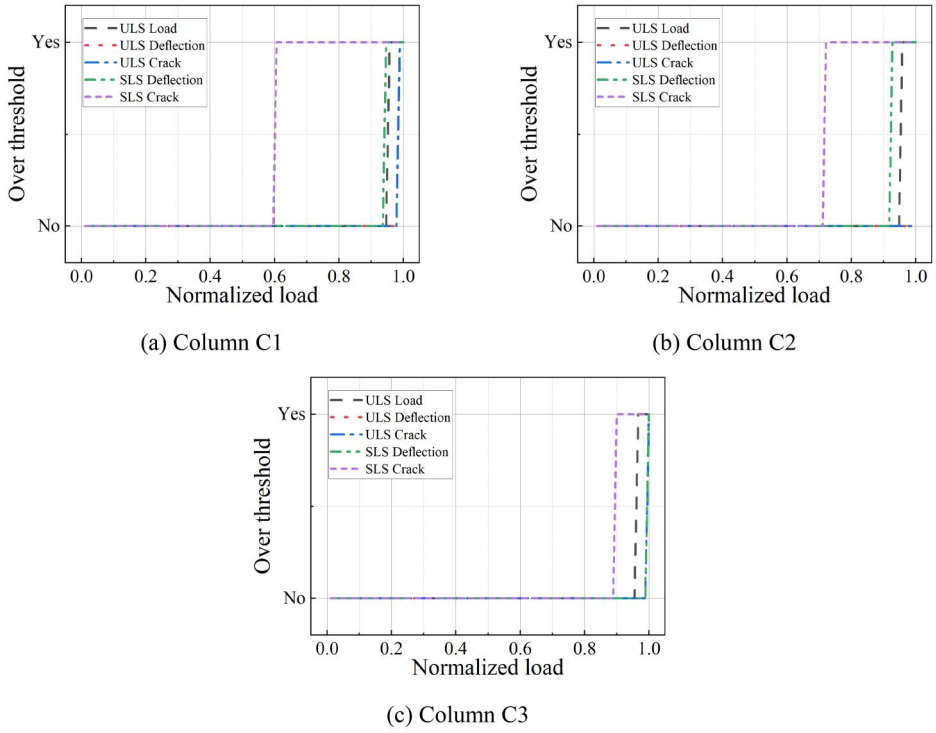


Figure 19. The state of four metrics for three RC columns at different loading levels.

Table 6. Condition metrics for quantitative assessment of RC columns.

	SLS			ULS			
Level	Definition	Crack (mm)	Deflection (mm)	Definition	Crack (mm)	Deflection (mm)	Load (kN)
L1	None defects	[0.00, 0.05)	[0%, 25%)· L_o/200	Safety	[0.00, 0.12)	[0%, 25%)· L_o/120	[0%, 25%)· R/0.95γ
L2	Slight defects	[0.05, 0.10)	[25%, 50%)· L_o/200	Low risk	[0.12, 0.25)	[25%, 50%)· L_o/120	[25%, 50%)· R/0.95γ
L3	Moderate defects	[0.10, 0.15)	[50%, 75%)· L_o/200	Medium risk	[0.25, 0.37)	[50%, 75%) · L_o/120	[50%, 75%)· R/0.95γ
L4	Significant defects	[0.15, 0.20)	[75%, 100%)· L_o/200	High risk	[0.37, 0.50)	[75%, 100%)· L_o/120	[75%, 100%)· R/0.95γ
L5	Severe defects	[0.20, $+\infty$)	[100%, $+\infty$)· L_o/200	Disaster risk	[0.50, $+\infty$)	[100%, $+\infty$) · L_o/120	[100%, $+\infty$)· R/0.95γ

7.2. Multi-metrics condition assessment method

The Dempster-Shafer (DS) evidence theory, proposed by Dempster and further developed by Shafer, has been widely used in the field of multi-source information fusion (Zhou et al., 2021). Sun et al. (2021) assessed the conditions of stay cables using the evidence theory in terms of inspection data. Yang et al. (2019) used the evidence theory and fuzzy analytic hierarchy process to assess the box girder bridge based on inspection and monitoring data. Zhang et al. (2023) performed multi-resource fusion of test data for bridge deck assessment using the discrete wavelet transformation and evidence theory. In this section, the evidence theory is used to conduct multi-metrics condition assessment of eccentrically loaded RC columns.

The serviceability and safety of RC columns are classified into five levels according to Ministry of Transport of the People's Republic of China (2011) and JTG 2120-2020 (Ministry of Transport of the People's Republic of China, 2020). Thresholds for these five levels are defined through engineering experience, expert consultation and relevant literature (Wang & Elhag, 2006; Zhang et al., 2023). Table 6 summarises the definitions of serviceability and safety, along with the corresponding metric ranges. These definitions will be used in the performance assessment of RC columns based on evidence theory.

As an example, Figure 20 illustrates a diagram of the ULS assessment process using the DS evidence theory based on multi-metrics (evidence). Firstly, the identification frame $\Theta = \{L_1, L_2, L_3, L_4, L_5\}$ represents the five condition assessment levels of RC columns under either the SLS or ULS assessment, as defined in Table 6. It is important to note that all subsets of Θ are mutually exclusive in this study. Therefore, the basic probability assignment (BPA) function, denoted as $m(A)$, is a mapping from the power set of Θ to the interval [0, 1] and satisfies the following conditions:

$$m(\varphi) = 0, \sum_{A \subset \Theta} m(A) = 1 \tag{18}$$

where A is a subset of the identification framework Θ; $m(A)$ is the BPA representing the degree of belief that supports A; and φ denotes the empty set within Θ.

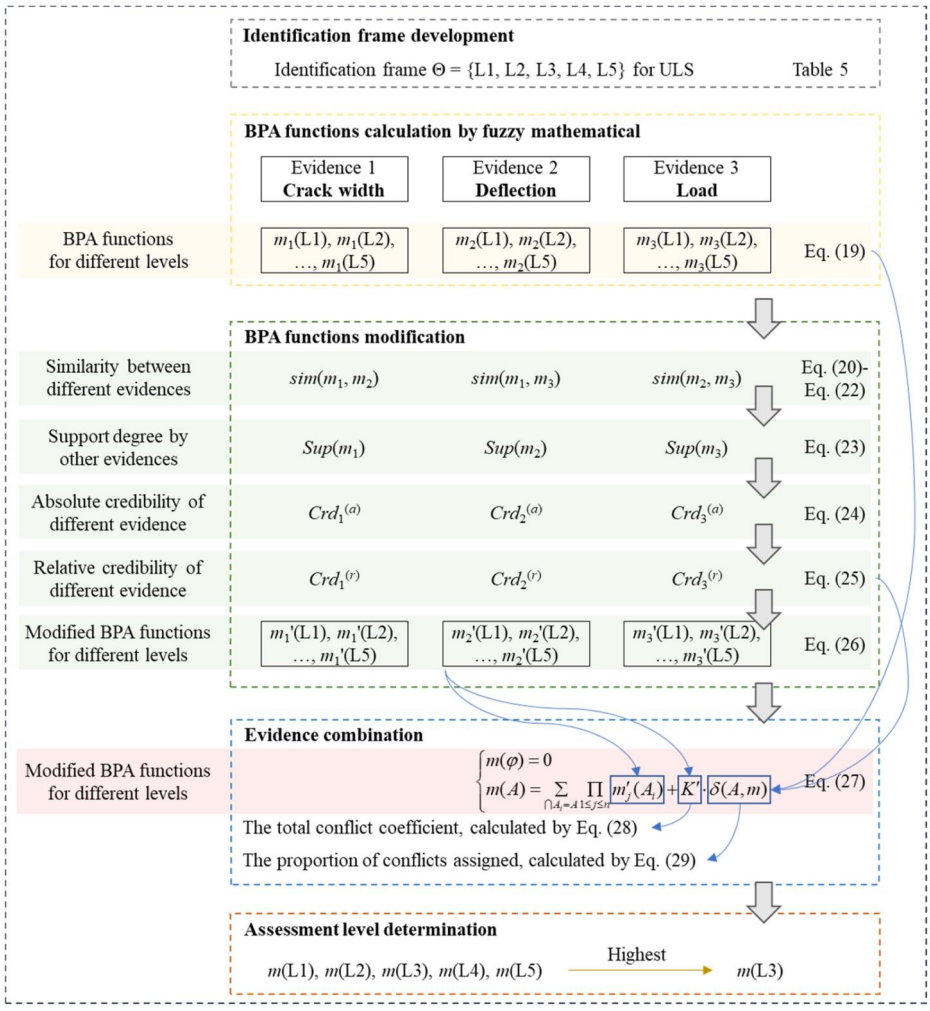


Figure 20. A diagram of evidence theory for multi-metrics ULS assessment.

In this study, two metrics (crack and deflection) are used for the SLS assessment and three metrics (crack, deflection and load) for the ULS assessment. The BPA function is determined based on the membership of the five condition levels for each metric, which is obtained using a fuzzy mathematical approach (Shu et al., 2024). The membership values are calculated using triangular fuzzy numbers (Sasmal & Ramanjaneyulu, 2008), as follows:

$$u_{ij} = \begin{cases} \dfrac{x_i - b_{ij}}{v_{ij} - b_{ij}}, & x_i \in [b_{ij}, v_{ij}]; \\ \dfrac{x_i - t_{ij}}{v_{ij} - t_{ij}}, & x_i \in [v_{ij}, t_{ij}]; \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

where u_{ij} is the membership of metric i to level j (L1–L5); x_i is the value of metric i obtained from the digital twin; v_{ij} denotes the mode of the triangular fuzzy number, corresponding to the left boundary of metric i for level j (as shown in Table 6).

The lower limit of the triangular fuzzy number b_{ij} is calculated by reducing the left boundary by 50% of the threshold, with a minimum value of 0. The upper limit of the triangular fuzzy number t_{ij} is determined by increasing the left boundary by 50% of the threshold, capped at the threshold value. Note that the 'crack width' and 'deflection' refer to the maximum crack width and mid-span deflection, respectively, derived from the digital twin. Figure 21 illustrates the triangular fuzzy numbers for the five levels. After computing the membership degrees, BPA functions for each evidence are obtained by summation normalisation. It should be noted that only five possible subsets (L1–L5) can be obtained by fuzzy numbers in the identification framework Θ.

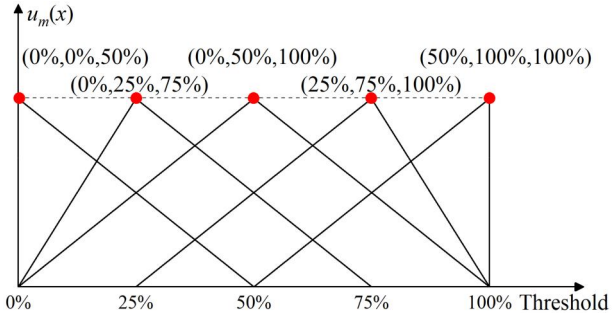


Figure 21. Fuzzy membership functions for the five fuzzy rating sets.

Then, conflicts among different pieces of evidence are resolved through evidence combination (Guo & Li, 2011).

Prior to this process, BPA functions are adjusted according to the credibility of each evidence, which is calculated using:

$$d(m_1, m_2) = \sqrt{(M_1 - M_2)^{\mathrm{T}} \boldsymbol{D} (M_1 - M_2)/2} \tag{20}$$

$$M_i = \begin{bmatrix} m_i(L1) & m_i(L2) & m_i(L3) & m_i(L4) & m_i(L5) \end{bmatrix} \tag{21}$$

$$sim(m_1, m_2) = 1 - d(m_1, m_2) \tag{22}$$

$$Sup(m_i) = \sum_{j=1, j\neq i}^{n} sim(m_i, m_j), i = 1, 2, \ldots, n \tag{23}$$

$$Crd_i^{(a)} = \frac{Sup(m_i)}{\max_{1\leq j\leq n}\left[Sup(m_j)\right]}, i = 1, 2, \ldots, n \tag{24}$$

$$Crd_i^{(r)} = \frac{Crd_i^{(a)}}{\sum_{j=1}^{n} Crd_j^{(a)}}, i = 1, 2, \ldots, n \tag{25}$$

where $d(m_1, m_2)$ denotes the distance between evidence 1 and evidence 2; $sim(m_1, m_2)$ represents their similarity; $\boldsymbol{D} = (D_{ij})$ is a $2^5 \times 2^5$ matrix, where $D_{ij} = |A_i \cup A_j|/|A_i \cap A_j|$ for i, $j = 1,2, \ldots, 2^5$; M_i is the BPA function of evidence i; $Sup(m_i)$ is the support degree of evidence i from other evidence sources; $Crd_i^{(a)}$ and $Crd_i^{(r)}$ represent the absolute and relative credibility of evidence i, respectively.

A smaller distance $d(m_1, m_2)$ implies higher similarity $sim(m_1, m_2)$, indicating stronger mutual support between the evidence sources. Generally, the more evidence is supported by others, the more credible it is considered. Once the support degree $Sup(m_i)$ is obtained, the absolute credibility $Crd_i^{(a)}$ is computed and normalised to yield the relative credibility $Crd_i^{(r)}$. The modified BPA function $m_i'(A)$ is then adjusted based on the absolute credibility, as defined in Equation (26). Finally, the modified BPA functions m_i' are combined using the relative credibility, as shown in Equations (27)–(29):

$$m'_i(A) = Crd_i^{(a)} \cdot m_i(A) \tag{26}$$

$$\begin{cases} m(\varphi) = 0 \\ m(A) = \sum_{\cap A_i = A} \prod_{1\leq j\leq n} m'_j(A_i) + K' \cdot \delta(A, m) \end{cases} \tag{27}$$

$$K' = \sum_{\cap A_i = \phi} \prod_{1\leq j\leq n} m'_j(A_i) \tag{28}$$

$$\delta(A, m) = \sum_{i=1}^{n} Crd_i^{(r)} \cdot m_i(A) \tag{29}$$

where K' represents the total conflict coefficient of the modified evidence model; and $\delta(A, m)$ determines the proportion of conflict allocated to the identification framework. Finally, the assessment level L for either the SLS or ULS is determined by selecting the subset with the highest probability in the identification frame, as expressed below:

$$\mathrm{L} = \max\left[m(\mathrm{L1})\ m(\mathrm{L2})\ m(\mathrm{L3})\ m(\mathrm{L4})\ m(\mathrm{L5})\right] \tag{30}$$

7.3. SLS and ULS assessment results

In this study, the ULS assessment of Column C1 under a 429 kN load is presented as a case study to illustrate the proposed computational procedure. The digital twin provides results for three ULS-related metrics: applied load, mid-span deflection and maximum crack width, which are 429 kN, 6.85 and 0.24 mm, respectively. First, the identification frame is defined as $\Theta = \{\mathrm{L1}, \mathrm{L2}, \mathrm{L3}, \mathrm{L4}, \mathrm{L5}\}$, corresponding to the five ULS levels specified in Table 7. The BPA functions for the three metrics are calculated using Equation (19), and the results are provided in Table 8. The distances between the evidence sources are then computed using Equation (20), giving $d(m_1, m_2) = 0.1362$, $d(m_2, m_3) = 0.7420$ and $d(m_1, m_3) = 0.664$. The small distance between the load and deflection metrics indicates a high degree of consistency in the ULS assessment.

Table 7. The BPAs for different metrics calculated by fuzzy mathematical.

		BPAs				
Metrics		Level 1	Level 2	Level 3	Level 4	Level 5
1	Load	0.9251	0.0499	0.0250	0.0000	0.0000
2	Deflection	0.1262	0.3762	0.3738	0.1238	0.0000
3	The maximum crack width	0.0298	0.2798	0.4702	0.2202	0.0000

Table 8. The Modified BPAs for different metrics.

		Modified BPAs				
Metrics		Level 1	Level 2	Level 3	Level 4	Level 5
1	Load	0.9894	0.0534	0.0267	0.0000	0.0000
2	Deflection	0.1179	0.3517	0.3496	0.1158	0.0000
3	The maximum crack width	0.0148	0.1385	0.2328	0.1090	0.0000

Based on Equations (22)–(24), the absolute credibility values of the three metrics are calculated as 1.07, 0.94 and 0.49. It is worth noting that the absolute reliability of the maximum crack width metric is only 0.49, which is lower than the other two metrics of 1.07 and 0.94. This means that there is a conflict between the three assessment results. The BPA values are subsequently adjusted using Equation (26), with the modified results presented in Table 8. The results show that the maximum crack width of BPA is significantly reduced, which can alleviate the conflict between the three metrics. The total conflict coefficient of the modified evidence is 0.9935. The evidence combination is then performed using Equation (27), resulting in the final ranking probability vector: [0.2485, 0.2764, 0.3409, 0.1345, 0.0000]. Among these, Level 3 has the highest probability value of 0.3409. Therefore, the ULS level of Column C1 under the applied load of 429 kN is determined to be Level 3.

Figure 22(a) shows the SLS and ULS condition assessment results for Column C1 under different load levels, following the method described in Section 7.2. The assessment metrics are obtained from the digital twin. The SLS level reaches 5 when the maximum crack width and deflection are 0.18 and 4.81 mm, respectively. Although these values do not exceed the defined thresholds, the DS evidence theory accounts for the collective uncertainty, resulting in a higher assessment level from a conservative viewpoint.

In contrast, the ULS assessment shows a delayed response compared to the SLS. When the ULS level reaches 5, the corresponding maximum crack width, deflection and load are 0.39 mm, 12.59 mm and 600.60 kN, respectively, all of which remain within the threshold values. This outcome highlights the DS theory's capacity to incorporate multiple metrics and better reflect the true condition of the structure.

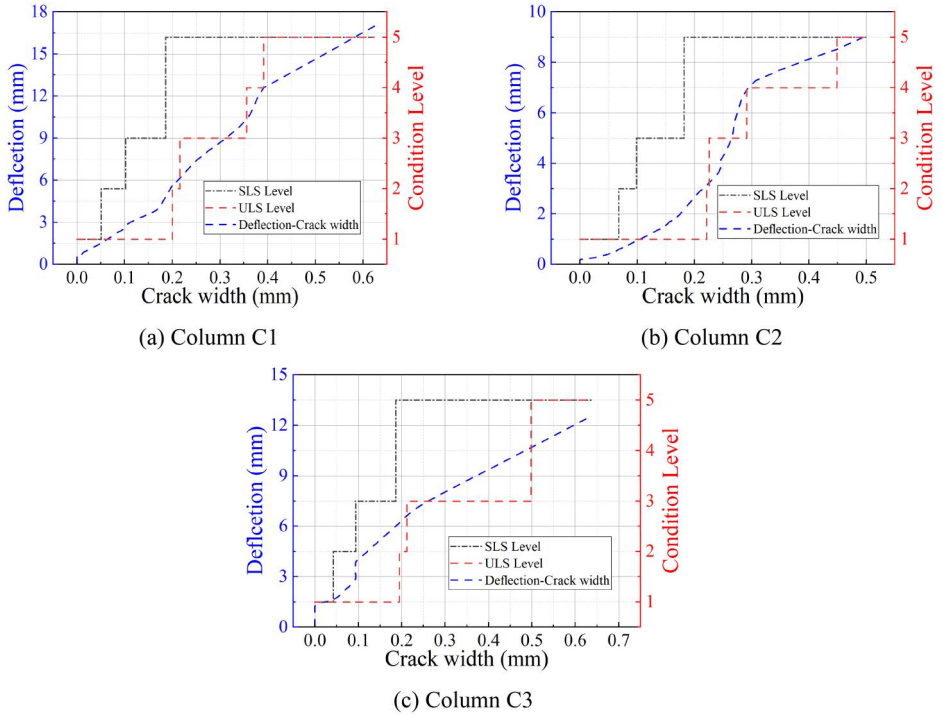


Figure 22. SLS and ULS assessments and condition levels for columns C1–C3.

Figure 22(b,c) shows similar assessment results for Columns C2 and C3. Compared with conventional threshold-based methods, the proposed DS-based assessment method demonstrates greater sensitivity and can effectively address inconsistencies between different metrics. Moreover, the use of digital twin-derived metrics improves the timeliness and continuity of structural condition evaluation.

8. Conclusions

The digital twin-driven multi-metrics condition assessment method has been developed, validated and applied in this study for eccentrically loaded RC columns with cracks. The main conclusions which can be drawn from this study are as follows:

1. The digital twin of eccentrically loaded RC columns with cracks can be developed effectively. The interaction between the physical and virtual entities is established using geometric information, deflections, concrete strains and maximum crack width, with implementation based on the PSO algorithm.
2. A methodology integrating 3D laser scanning and rebar scanning is developed to capture column geometry and reinforcement layout and reduce geometric uncertainties, enabling high-quality gDTs. This provides a reliable foundation for more accurate inference of mechanical parameters through model updating.
3. The accuracy of the developed mechanical digital twin is validated by comparing its predicted responses with experimental data, including concrete spalling, tensile and lateral concrete strains and reinforcement strains. The identified material parameters can be extended to predict the behaviour of other RC columns fabricated under similar conditions.
4. An improved condition assessment method based on DS evidence theory is proposed, which effectively addresses conflicts among multiple metrics. Compared to conventional threshold-based approaches, the proposed method demonstrates higher sensitivity and enhances the overall reliability of the assessment.
5. The digital twin framework enables more timely condition assessment. A functional relationship between deflection and maximum crack width is derived from the digital twin, allowing crack width estimation from monitored deflection data. This demonstrates the digital twin's reliability and effectiveness in structural condition evaluation.

With recent technological advancements, point cloud and rebar scanning techniques provide efficient support for structural condition assessment. However, these methods face limitations in severely occluded areas or where direct detection is difficult. Addressing data gaps and geometric occlusion remains a priority. Furthermore, the digital twin developed in this study was established and validated without considering environmental degradation. Future work should examine the framework under various loading conditions, environmental actions and different cross-sectional forms, to enhance its generalisability and predictive capability for practice. Its applicability to in-service RC beams with initial defects or damage also merits investigation.

List of symbols

e_0	Loading eccentricity of columns (mm)
E_c	Young's modulus of concrete (MPa)
f_c	Compressive strength of concrete (MPa)

f_t	Tensile strength of concrete (MPa)
ν	Poisson's ratio (/)
σ_c	Compressive stress of concrete (MPa)
σ_t	Tensile stress of concrete (MPa)
d_c	The damage variable for compressive of concrete (/)
d_t	The damage variable for tensile of concrete (/)
ε_c	The peak compressive strain of concrete (/)
ε_t	The peak tensile strain of concrete (/)
α_c	The parameter of the descending section of the uniaxial compressive stress–strain curve of concrete (/)
α_t	The parameter of the descending section of the uniaxial tensile stress–strain curve of concrete (/)
φ	The dilation angle of concrete (°)
e	The flow potential eccentricity of concrete (/)
E_s	Young's modulus of reinforcement (MPa)
f_y	Yield strength of reinforcement (MPa)
f_u	Ultimate strength of reinforcement (MPa)
σ^s	Compressive stress of reinforcement (MPa)
ε^s	Tensile stress of reinforcement (/)
E_i	The slope of the reinforcement after yielding (/)
$\tau_{b\max}$	The peak bond shear stress between concrete and reinforcement (MPa)
τ_{bf}	The residual bond shear stress (MPa)
ε_e	The elastic strain (/)
ε_f	The crack strain (/)
w_f	The crack width (mm)
l_E	The equivalent length of the element (mm)
P_u	The ultimate load-bearing capacity (kN)

Author contributions

CRediT: **Kai-Wen Zhang**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft; **You-Lin Xu**: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Writing – review & editing; **You Dong**: Resources, Supervision; **Zhong-Cheng Zeng**: Data curation, Investigation; **Lian-Heng Cai**: Investigation, Writing – review & editing; **Yu-Hang Ding**: Data curation, Investigation; **Yong-Ding Tian**: Resources, Supervision.

Disclosure statement

The authors declare that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

All data, models, or codes that support the findings of this study are available upon reasonable request.

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Appendix A

Algorithm 1. Geometry vertices calculation

Input: point cloud coordinates $P_{n\times 3}$
Output: 16 column coordinates $C_{n\times 16}$
For $i = 1$ to 16 **do** // *Traverse all column vertices*
  // *For each column, fit 3 planes to determine its coordinate*
  For $j = 1{:}3$ **do** // Fit 3 intersecting planes for each column
    S_j = *SelectPointsInteractively*(P) // *Interactive point selection on plane j*
    (a_j, b_j, c_j, d_j) = *FitPlaneLeastSquares*(S_j) // *Plane fitting:* $a_jx + b_jx + c_jx + d_jx = 0$
  end for
  // Solve the intersection point of 3 planes (column coordinate)
  $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ // *Coefficient matrix*
  $B = \begin{bmatrix} -d_1 \\ -d_2 \\ -d_3 \end{bmatrix}$ // *Constant vector*
  $C(i, :)$ = *SolveLinearSystem*(A, B)
end for

Appendix B

Algorithm 2. Material parameter optimisation based PSO in ABAQUS and Python

Input: Test results; Initial material parameters; population; Iterations n
Output: material parameters; fitness; response of column
Function main_optimisation()
  target_data ← *LOAD_TARGET_DATA*() // *Load test results*
  particles ← *INITIALIZE_POPULATION*() // *Initialisation population*
  For $i = 1{:}n$ // *Start the iteration process*
    For $j = 1$: *population* // *Submit the calculations for each population*
      RUN_ABAQUS(j, i, particles[j]) // *ABAQUS calculation*
    end for
    fitness_scores ← [] // *Fitness storage*
    For $k = 1$: *population* // *Traverse each population*
      results ← *EXTRACT_RESULTS*(k) // *Extract the results*
      fitness ← *CALCULATE_FITNESS*(*results*, *target_data*)
      fitness_scores.append(*fitness*)
  end for
  UPDATE_PARTICLE_SWARM(*particles*, *fitness_scores*) // Update
 end for
 Return *BEST_SOLUTION*
end function
// *Key Function*
Function *RUN_ABAQUS*(*job_id, generation, parameters*) // *run ABAQUS by python*
  UPDATE_MATERIAL_PROPERTIES(*parameters*)
  GENERATE_INPUT_FILE(*job_id*)
  SUBMIT_ANALYSIS(*job_id, generation*)
end function
Function *EXTRACT_RESULTS*(*job_id*) // *Extract the results from the ABAQUS odb*
  OPEN_ODB_FILE(job_id)
  Element_sets ← [*U*1, *U*2, *U*3, … , *U*5, *RF*] // *Required elements or nodes name*
 For $i = 1{:}$*Element_sets* // *Traverse each elements or nodes*
  EXTRACT_CRACK_WIDTH()
  EXTRACT_STRAIN_DATA()
  EXTRACT_DISPLACEMENT()
 end for
 return *compiled_results*
end function